

Benjamin Curis-Friedman

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EDUCATION

McGill University

Bachelor of Software Engineering COOP 3.96 GPA

Montreal, QC

Aug. 2024 – Dec. 2027

Awards (6000\$): Alma Mater Entrance Scholarship, Class of Engineering '83 Scholarship

Stanford Online

Machine Learning Specialization by Andrew Ng 100% Grade

May 2025 – August 2025

Covering regression, deep-learning, random forests, and ML engineering practices.

EXPERIENCE

Software Developer Intern

Jan. 2026 – Present

Rogue Research Inc

Montreal, QC

- Rendered realistic optical distortion by writing custom vertex shader replacements for vtk in OpenGL.
- Contributed to open source libraries (pyvista & vtk), improving rendering performance and code coverage.
- Engineered a 3D robot emulator application using pyvista, enabling accurate surgery planning.
- Refactored a brain surgery robot networking protocol to make it resilient to corrupt data packets.
- Implemented robotic transcranial magnetic stimulation algorithms, setting a baseline for further research.
- Developed unit tests for existing classes and methods to improve code coverage in Objective-C.
- Implemented a vectorized SSIM image comparison algorithm for CI UITests using Apple's vImage framework.

Artificial Intelligence Intern

May 2025 – Dec. 2025

iSMART AI Lab

Montreal, QC

- Developed a video synchronization & compression library using FFmpeg and OpenCV to prepare for training.
- Coded a data pre-processing pipeline to clean raw human vital signals for machine learning models.
- Built and programmed a task labeling device using Arduino and C++ to split recording segments appropriately
- Created a high quality multi-modal dataset of over 50 hours of recordings from dozens of study participants.

PROJECTS

Omniclave 🌀 | *FastMCP, FastAPI, Auth0, React, Bun*

May 2026

- Built an MCP server for a student portal, empowering agents to help with homework and emails.
- Integrated OAuth2 to let users login from their favourite MCP client (i.e. Codex / Claude code).
- Automated a secondary login flow using a backend playwright instance to log into Omnivox on a user's behalf.
- Reverse engineered the Omnivox website to replicate the http headers and intercept the bearer token.

AI Library From Scratch 🌀 | *Python, NumPy, TensorFlow, Keras, PyGame*

June 2025 – July 2025

- Created a NN library complete with mini-batch gradient descent and activation/cost functions.
- Trained a sequential neural network achieving ~96% test accuracy on MNIST dataset.
- Implemented customizable network and layer shapes for enhanced flexibility and scalability.
- Visualized inference with an interactive real-time digit recognition game using PyGame.

Cheese Manager 🌀 | *Java, JavaFX*

September 2025

- Developed an application for a comté cheese distribution business following the MVC pattern in Java.
- Created an aesthetically pleasing user interface with JavaFX using reusable components.
- Collaborated with teammates to create a UML class diagram and state diagram using Umple.

Web Chat Application 🌀 | *Firebase, Javascript, Vue.js, HTML/CSS*

Oct. 2022 – Sep. 2023

- Developed a full-stack web messaging platform with authentication to message friends.
- Programmed search and adding friends features with Cloud Functions.
- Implemented Firestore security rules to ensure secure communications.

SKILLS

Languages: Python, C, Objective-C, Java, Bash, C++, Typescript, JavaScript, HTML/CSS, R

Frameworks: React, Node.js, JUnit, Tailwind, Cocoa, JavaFX

Developer Tools: Git, Docker, CDash, VS Code, XCode, NeoVim, Linux, Subversion, Google Firebase

Libraries: NumPy, Matplotlib, PyVista, VTK, OpenCV, pandas, SciPy, Keras, TensorFlow

Spoken Languages: English, French